

Residual LSTM Forecasting for Multivariate Operational Time Series

Tudor Orza Marie Reinhold Felix Besser
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Abstract. We forecast hourly operational load for 96 anonymized units. The required system predicts 336 hours recursively in 24-hour blocks and is evaluated with WAPE. Our compact PyTorch LSTM predicts corrections to a strong four-week seasonal mean, using known future covariates and learned unit embeddings. On a chronological holdout it reaches WAPE **0.2413**, compared with 0.3418 for the seasonal mean and 0.5450 for last-value persistence. We also transfer the architecture to joint Jena temperature and humidity forecasting.

0.2413 internal WAPE - 29% below the strongest seasonal baseline

1. Introduction

Operational load forecasting supports staffing, capacity planning, and early recognition of system pressure. The released benchmark contains 96 related but distinct hourly series, 22 explanatory variables, and one nonnegative target. Future covariates are supplied, whereas future target values are hidden. The model must therefore combine daily and weekly regularities with unit-specific dynamics and external signals while limiting error accumulation over fourteen recursive blocks.

We favor a small recurrent model over a much larger architecture so controlled ablations and CPU-only evaluation remain practical. The work contributes a seasonally anchored residual LSTM, leakage-safe preprocessing, exact recursive inference, and a two-target transfer experiment on Jena climate data.

Problem contract

Frequency	History	Block	Required horizon	Metric
Hourly	168 h	24 h	336 h	WAPE

2. Related Work

Long short-term memory networks use gated recurrent updates to preserve useful information over long sequences [1]. Recurrent architectures remain widely used for nonlinear sequential modeling [2], including robust time-series prediction [3]. Attention-based recurrent models can separate feature selection and temporal attention [4], but the present data volume and reproducibility requirements favor a smaller model.

Seasonal persistence is unusually competitive in short-horizon forecasting. Instead of asking a neural network to reconstruct the entire weekly level, we use a seasonal mean as an explicit inductive bias and learn only deviations explained by current load, risk, capacity, and calendar variables.

3. Method

3.1 Leakage-safe preprocessing

Rows are sorted per series and checked for duplicate timestamps and hourly spacing. The final 336 public training hours are reserved for selection. Continuous covariates are standardized with statistics from the remaining interval only; binary and cyclic variables are unchanged. Missing inputs receive an indicator, are forward-filled within their series, and use the training median only if no earlier value exists.

A training window contains 168 history hours and 24 output hours. Starts are six hours apart and occur only after four complete weeks. Historical targets are standardized per series for the encoder, while outputs and the L1 loss remain on the original scale.

3.2 Architecture

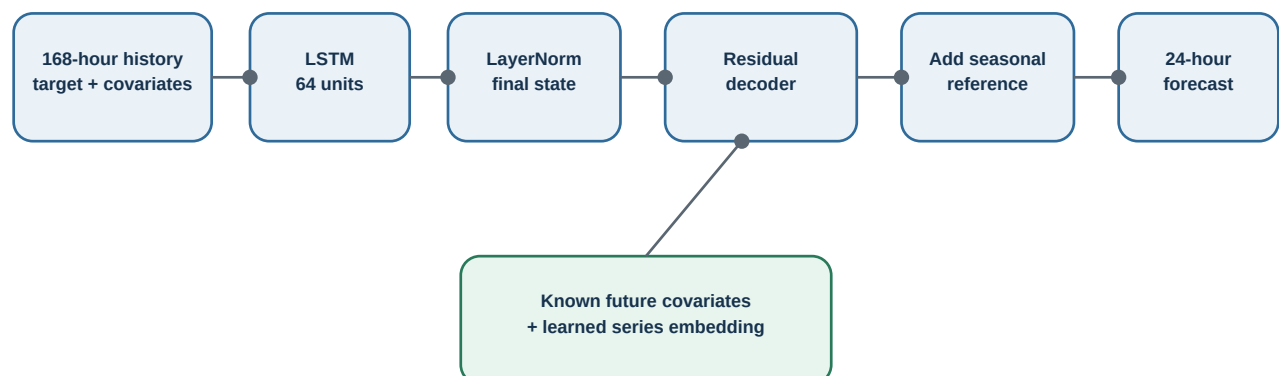


Figure 1. The encoder state, future covariates, and unit embedding drive a shared residual decoder. The residual is added to the seasonal reference.

3.3 Optimization and recursive inference

The encoder has one LSTM layer with 64 hidden units. Its final state is processed by Layer Normalization and concatenated with an eight-dimensional series embedding and known future covariates. A two-layer decoder emits 24 residuals. The reference averages matching hours from the previous four weeks, falling back to weekly and then last-value persistence when history is short. Predictions are clamped to zero only during inference.

We optimize raw-scale L1 with AdamW, learning rate 0.001, weight decay 0.0001, batch size 256, and gradient clipping at 1.0. Training uses seed 42, at most 30 epochs, and patience five. Each predicted block is appended to history before the next one, exactly matching private evaluation.

4. Benchmark Experiments

4.1 Protocol

The internal holdout contains the last 336 hours for every unit. No held-out target is exposed during preprocessing, training, or rollout. WAPE is aggregated across all units and timestamps. Once the full model was selected, it was retrained from initialization on all public targets for the selected 12 epochs.

Method	Internal WAPE	Compared with last value
Last value	0.545040	reference
Daily persistence	0.418006	-23.3%
Weekly persistence	0.454876	-16.5%
Four-week seasonal mean	0.341780	-37.3%
Target-history-only LSTM	0.383264	-29.7%
Full multivariate LSTM	0.241293	-55.7%

Table 1. Chronological benchmark results; lower WAPE is better.

4.2 Ablation and interpretation

The target-only LSTM beats last-value, daily, and weekly persistence but not the four-week seasonal mean. The full model reduces WAPE by 29.4% relative to that strongest baseline. Since both neural variants share the seasonal anchor, encoder, decoder, and optimizer, the difference is direct evidence that the released covariates provide information beyond target history. Validation improved through epoch 12; later training-loss reductions did not generalize.

The final CSV contains exactly 32,256 finite, nonnegative predictions in forecast index order. The packed checkpoint stores feature ordering, imputation statistics, scalars, series mapping, recent context, architecture settings, and weights.

5. Additional Dataset: Jena Weather

The Jena climate record from the Max Planck Institute for Biogeochemistry contains ten-minute meteorological observations from 2009 through 2016 [5]. We aggregate numerical variables to hourly means. Negative sentinel wind speeds are marked missing before aggregation, and wind direction is converted to Cartesian x/y components. Air temperature and relative humidity are predicted jointly.

5.1 Experimental design

Calendar years 2009-2014 form the training split, 2015 is validation, and 2016 is test. At each daily origin, 168 observed hours predict the next 24. Historical meteorological measurements enter the encoder; only hour- and year-cycle sine/cosine features are available for future timestamps. This prevents access to future measured weather. The model selected epoch 12 by mean normalized validation MAE.

Method	Temp. MAE	Temp. RMSE	Humidity MAE	Humidity RMSE	Norm. MAE
Last value	2.719	3.712	11.169	15.126	0.501
Daily persistence	2.432	3.202	9.884	12.991	0.445
Weekly persistence	4.565	5.704	12.157	15.522	0.639
Residual LSTM	2.706	3.442	8.903	11.745	0.431

Table 2. Jena 2016 test results across 366 non-overlapping forecast origins.

5.2 Transfer analysis

The LSTM has the best combined normalized MAE and improves humidity MAE by 10% relative to daily persistence. Daily persistence remains 0.275 degrees C better for temperature MAE, showing that no single method dominates each channel. The model transfers successfully at the architecture level: only the number of series, output channels, historical measurements, and future calendar inputs change.

Weather also exposes a limitation of a fixed seasonal anchor. Temperature has a strong daily cycle, while humidity reacts to fronts and precipitation that are not known at the forecast origin. The multivariate encoder helps humidity most, but a future experiment could add numerical weather predictions or target-specific loss weights.

Jena result: best combined error and best humidity forecast

6. Relation to the Expose

The final method follows the expose's selection of an LSTM and its planned daily and weekly baselines. Inspection of the released data required several corrections. The benchmark is operational rather than meteorological, so its known load, risk, capacity, and calendar covariates are modeled directly. Jena measurements occur every ten minutes rather than every twenty. We replaced the tentative percentage split with reproducible calendar-year boundaries and made the weather objective explicit: joint temperature and humidity prediction. AR, MA, and ARMA remain related comparison ideas rather than implemented models.

7. Limitations

The benchmark analysis uses one temporal holdout and one initialization, so it does not measure variance across cutoffs or random seeds. Recursive errors may grow under distribution shift, and the seasonal reference assumes stable weekly structure. The compact LSTM cannot represent every long-range interaction available to larger attention or state-space models. The Jena experiment uses observed measurements only in history, making rapidly changing weather inherently difficult.

8. Conclusion

A small residual LSTM substantially outperforms strong persistence rules on the operational benchmark while remaining easy to reproduce and package. The target-only ablation demonstrates that the main gain depends on multivariate information rather than neural capacity alone. On Jena, the architecture achieves the best combined temperature/humidity error and materially improves humidity forecasting. The final archive runs offline on CPU and reproduces the evaluator's exact command.

Contributions

Required author check before submission: Tudor Orza, Marie Reinhold, and Felix Besser must replace this note with their agreed individual contributions. No allocation was invented because it cannot be inferred reliably from the code or expose.

Reproducibility checklist

Artifact	Status
Deterministic training configuration	Complete
Benchmark and Jena result JSON	Complete
Exact CPU inference interface	Verified
Self-contained model archive	Verified
Hugging Face validation score	Pending external submission
Individual contribution statement	Pending group confirmation

References

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The report body occupies five pages; this reference page is excluded from the assigned 4-6 page limit.